

January 2026

IPC Online

Section: B2

Time: 30 min

Run three threads.

- Thread1 prints 'P' in an infinite loop
- Thread2 prints 'Q' in an infinite loop
- Thread3 prints 'R' in an infinite loop

Now, these three threads should run independently and should print only one letter at a time. At each iteration, 'P', 'Q' and 'R' - all three letters must be printed in random order. Also the number of each letter to be printed at each iteration equals the iteration count . So, the output may look like.

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Implement this using semaphore only. Take iteration count as input.

Input format: <Number of iterations>

Output:

PQR	[iteration 1]
QQPRRP	[iteration 2]
PQPQRQRRP	[iteration 3]

Note that the letters can be printed in any order.

Create a file <student_id>.cpp and submit it.